

Sr. No.		Description	UOM (Wherever Applicable)	Data (Common For All Models)
A		General Points		
	1	Cooling Capacity	ton _R	Refer KCPL Chiller Selection System Software
	2	Power Consumption	kW	Refer KCPL Chiller Selection System Software
	3	Specific Power Consumption	kW/ton _R	Refer KCPL Chiller Selection System Software
	4	Co-Efficient of Performance (COP)	kW/kW	Refer KCPL Chiller Selection System Software
	5	No. of Compressors	Nos.	Contact To Engineering Team
	6	No. of Individual Refrigerant Circuits	Nos.	Contact To Engineering Team
	7	Refrigerant		
	i	Name	-	R134a
	ii	Quantity	kg	Refer KCPL Chiller Selection System Software
	iii	Technical Specifications	-	Refer ESP-18-19-003
	8	Sound Pressure Level		
	i	Noise Level	dB	Refer ESP-18-19-001
	ii	Measuring Standard	-	ANSI/AHRI Standard 575-2008
	9	Insulation Details		
	i	Material	-	Closed Cell Nitrile Foam
	ii	Insulation Thickness on Various Parts	-	For Standard Temperature Range (LWT upto -10 °C)
		Evaporator Shell	mm	32
		Evaporator Tubesheet	mm	19
		Evaporator Pass Partition Assembly	mm	19
		Evaporator Head Cover	mm	32
		Evaporator Support Plate	mm	19
		Compressor Motor Body	mm	19
		Suction Line Assembly	mm	19
		Liquid Line Assembly	mm	9
	iii	Insulation Thickness on Various Parts	-	For Brine Temperature Range (LWT below -10 °C)
		Evaporator Shell	mm	51 (32+19)
		Evaporator Tubesheet	mm	32
		Evaporator Pass Partition Assembly	mm	32
		Evaporator Head Cover	mm	51 (32+19)
		Evaporator Support Plate	mm	32
		Compressor Motor Body	mm	28 (19+9)
		Suction Line Assembly	mm	28 (19+9)
		Liquid Line Assembly	mm	19
	iv	Density	kg/m ³	76.6
	v	Thermal Conductivity	W/m.K	0.035 (at 0 °C Mean Temperature)
	vi	Standard	-	IS 14164
	vii	Adhesive	-	Blend of Synthetic Polymers and Synthetic Resin
	viii	Insulation Specifications	-	Refer ESP-18-19-004
	10	Vibration		
	i	Vibration Level	mm/sec	Less than 1.5 mm/sec
	ii	Vibration control	-	Rubber Pads (Standard) / Spring Isolators (At an Additional Cost)
	iii	Standard	-	IS 12075
	11	Painting Specification		
	i	Paint Type	-	RAL 7035
	ii	Standard	-	Coating as per KCPL Standards
	12	Overall Dimensions		
	i	Approx. Length	mm	Refer KCPL Chiller Selection System Software
	ii	Approx. Width	mm	Refer KCPL Chiller Selection System Software
	iii	Approx. Height	mm	Refer KCPL Chiller Selection System Software
	13	Space Clearances Required		
	i	Panel Side	mm	Contact To Engineering Team
	ii	Opposite to Panel Side	mm	Contact To Engineering Team
	iii	All Other Sides	mm	Contact To Engineering Team
	iv	Overhead	mm	Contact To Engineering Team
	14	Weight		
	i	Approx. Shipping Weight	kg	Refer KCPL Chiller Selection System Software
	ii	Approx. Operating Weight	kg	Refer KCPL Chiller Selection System Software
	15	Cable Sizes		
	i	Aluminum Cable	-	Refer ESP-14-15-01
	ii	Copper Cable	-	Refer ESP-14-15-01
B		Compressor Details		
	1	Make	-	Kirloskar Chillers Private Limited
	2	Type / Description	-	Semi-Hermetic Twin Screw Compressor
	3	Model	-	Refer KCPL Chiller Selection System Software
	4	Drive	-	Direct Driven by Rotor Shaft
	5	Capacity Control Percentage	%	Contact To Engineering Team
	6	Type of Capacity Control	-	Stepless
	7	Capacity Control Mechanism	-	Slide Valve Mechanism

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	8		Volumetric Ratio	-	Fixed Ratio (3.2)
	9		Design and Test Parameters		
		i	Design Pressure	bar	30
		ii	Test Pressure (Pneumatic)	bar	33
		iii	Design Temperature	°C	120
		iv	Max. Allowable Discharge Temperature	°C	120
	10		Bearings		
		i	Types of Bearings	-	Roller Bearings - For Radial Load
					Angular Contact Roller Bearing - For Axial Load
		ii	Material of Construction	-	Steel

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		iii	Life of Bearing	Hours	50000
		iv	Class of Bearing	-	Proprietary Data
	11		Lubrication		
		i	Type	-	Lubrication by Differential Pressure Mechanism
		ii	Lubricating Oil	-	Synthetic Oil
		iii	Grade of Lubricating Oil	-	Proprietary Data
		iv	Quantity	Liter	Refer KCPL Chiller Selection System Software
	12		Compressor Components MOC		
		i	Screw	-	Alloy Steel
		ii	Casing	-	Cast Iron
		iii	Shaft	-	Alloy Steel
		iv	Rotor	-	Aluminum Alloy
	13		Physical Data of Compressor		
		i	Screw Construction	-	Twin Screw
		ii	No. of Lobes Male Rotor	Nos.	5
		iii	No. of Lobes Female Rotor	Nos.	6
		iv	Male Rotor Diameter (mm)	mm	Proprietary Data
		v	Female Rotor Diameter (mm)	mm	Proprietary Data
		vi	Driving Rotor	-	Male Rotor
	14		Oil Filter		
		i	Micron Rating	Micron	10
		ii	Material of Construction	-	Resin Impregnated Fibres
		iii	Quantity	Nos.	1 No. per Compressor
	15		Copressor Isolation Type		
		i	At Suction	-	Butterfly Valve
		ii	At Discharge	-	Shut-off Valve
C			Compressor Motor Details		
	1		Make	-	Kirloskar Approved Vendor
	2		Motor Type	-	Semi-Hermetic Squirrel Cage Induction Motor
	3		Type of Duty	-	Continuous
	4		Motor Rating	kW	Refer KCPL Chiller Selection System Software
	5		Motor Speed (Synchronous)	RPM	3000
	6		Ingress Protection (IP)	-	NA, Being Semi-Hermetic Type
	7		GD ² of Rotor	-	Proprietary Data
	8		Whether SPDP or TEFC?	-	NA, Being Semi-Hermetic Type
	9		Power Supply Details (Standard)		
		i	Supply Voltage	V	400
		ii	Permissible Voltage Variation	%	±10%
		iii	Frequency	Hz	50
		iv	Permissible Frequency Variation	%	±3%
		v	Phase	-	3
	10		Performance Indicators		
		i	Motor Efficiency Class	-	NA
		ii	Motor Power	kW	Refer KCPL Chiller Selection System Software
		iii	Motor Efficiency	-	Consult with Engineering Department on Case to Case Basis
		iv	Power Factor	-	Consult with Engineering Department on Case to Case Basis
		v	Class of Insulation	-	Class F
	11		Motor Cooling		
		i	Motor Cooling Type	-	Refrigerant Cooled
		ii	Cooling Mechanism	-	Suction Gas
		iii	Temperature at full load	⁰ C	10 to 15 (At Normal Condtns)
	12		Current Details		
		ii	Rated Load Current	A	Refer KCPL Chiller Selection System Software
		iii	Full Load Current	A	Refer KCPL Chiller Selection System Software
		iv	Inrush/Starting Current	A	Refer KCPL Chiller Selection System Software
		v	Locked Rotor Current	A	Refer KCPL Chiller Selection System Software
		vi	Starting Torque	N.m	Contact To Engineering Team
		vii	No Load Current	A	Contact To Engineering Team
		viii	Acceleration Time to Reach Rated Speed	Sec	2 to 3
	13		Control Settings		
		i	No. of Starts per Hour	Nos.	4
		ii	Time Between STOP to START	Sec	300
		iii	Time Between START to START	Sec	900
D			Power Supply (Standard-Chiller Icomer)		
	1		Supply Voltage	V	415
	2		Permissible Voltage Variation	%	±10%
	3		Frequency	Hz	50
	4		Permissible Frequency Variation	%	±3%

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	5		Phase	-	3
	6		Control Voltage	V	230 (Standard)
					110 (Special-Optional)
	7		Supply Wire System	-	3 Phase - 4 Wire System (Standard)
					3 Phase - 3 Wire System (Special-Optional)
	8		Fault Level at Busbar	kA	As per KCPL Standard Practice
E			Oil Separator Details		
	1		Type	-	Dome Type (Built in Compressor)
	2		Internal Structure	-	Demister Arrangement
	3		Method of Oil Separation	-	Differential Mass Between Oil and Gas, Impact with Surfaces, Filtering of Oil-Gas Mixture
	4		Oil Heater Details		
		i	Make	-	Kirloskar Approved Vendor
		ii	Quantity	Nos.	Contact To Engineering Team
		iii	Power Supply	V	230
		iv	Rating	W	250

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F		Oil Cooler	-	If Applicable
	1	Type	-	Plate Type
	2	Quantity	Nos.	One per Compressor
	3	Heat Duty	kW	Depends on Working Conditions
	4	Method of Cooling	-	Refrigerant Cooled
	5	Material of Construction	-	Brazed PHE, Plate Material - SS
	6	Pressure Drop		
	i	Oil Side	bar	less than 0.5
	ii	Refrigerant Side	bar	Proprietary Data
G		Evaporator Details		
	1	Model	-	Refer KCPL Chiller Selection System Software
	2	Design Code	-	As per KCPL Standards
	3	Type	-	Shell and Tube DX Design
	4	Tube Side (Fluid)	-	Refrigerant
	5	Shell Side (Fluid)	-	Chilled Water
	6	Design Parameters		
	i	Design Temperature (Refrigerant Side)	⁰ C	65
	ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
	vi	No. of Passes (Refrigerant Side)	Nos.	Contact To Engineering Team
	vii	Design Temperature (Water Side)	⁰ C	65
	viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
	ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
	x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
	xi	Testing method (Water Side)	-	Refer ESP-07-08-107
	xii	No. of Passes (Water Side)	Nos.	Single Pass
	xiii	Water Velocity	m/s	Less than 3 m/s
	xiv	Inlet Pressure	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
	xv	Evaporating Temperature	⁰ C	Consult with Engineering Department on Case to Case Basis
	7	Physical Data of Evaporator		
	i	Overall Length of Evaporator	ft	Contact To Engineering Team
	ii	Shell Diameter	inch	Contact To Engineering Team
	iii	Shell Thickness	mm	Contact To Engineering Team
	iv	Approx. Shell Length	mm	Contact To Engineering Team
	v	Material of Construction of Shell	-	Mild Steel
	vi	Material Standard of Shell	-	Refer "MOC" Sheet
	vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
	viii	Tube Length	mm	Refer "HX Details" Sheet
	ix	Tube Diameter	mm	Refer "HX Details" Sheet
	x	Tube Thickness	mm	Refer "HX Details" Sheet
	xi	Material of Construction of Tube	-	Cu
	xii	Material Standard of Tube	-	Refer "MOC" Sheet
	xiii	Water Volume in Evaporator	Liter	Refer KCPL Chiller Selection System Software
	8	Water Box Details		
	i	Type	-	Standard - On Shell Nozzle
	ii	Material	-	Mild Steel
	iii	Standard (Material)	-	Refer "MOC" Sheet
	iv	Nozzle size	NB	Refer KCPL Chiller Selection System Software
	v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
	vi	MOC of Water Side Gasket	-	NAM AF 120
	vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
	9	Accessories Provided		
	i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)
	ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
H		Condenser Coil Details		
	1	Make	-	Kirloskar Approved Vendor
	2	Type	-	Fin and Tube Design
	3	Coil Arrangement	-	V' Type
	4	Tube Side (Fluid)	-	Refrigerant
	5	Fin Side (Fluid)	-	Air
	6	Design Parameters		
	i	Design Temperature (Refrigerant Side)	⁰ C	100
	ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107

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		vi	Total Heat Rejection	ton _R	Formula - THR = Chiller Cooling Capacity + (3.51685/Input Power)
		vii	Condensing Temperature	⁰ C	Consult with Engineering Department on Case to Case Basis
		viii	Degree of Subcooling	⁰ C	5
	6		Physical Structure		
		i	No. of Passes (Refrigerant Side)	Nos.	4
		ii	No. of Rows (Refrigerant Side)	Nos.	3
		iii	Coil Face Area	m ²	2.23875 per coil
		iv	Air Velocity	m/s	Average 3 m/s
		v	Air Flow Rate	m ³ /s	Refer KCPL Chiller Selection System Software
		vi	No. of Coils	Nos.	Refer KCPL Chiller Selection System Software
		vii	Fin Density (No. of Fins/inch)	Fins/inch	14

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	viii	Tube Diameter	mm	Refer "HX Details" Sheet
	ix	Tube Thickness	mm	Refer "HX Details" Sheet
	x	Material of Construction of Tubes	-	Cu
	xi	Material Standard	-	Refer "MOC" Sheet
	xii	Material of Construction of Fins	-	Aluminum
	xiii	Type of Coating	-	Standard - No Coating
				Hydrophilic Coating - Optional
				Blygold Coating - Optional
	xiv	Thickness of Layer (Coating)	mm	Consult with Engineering Department on Case to Case Basis
	xv	Life of Fins with Coating	Years	Consult with Engineering Department on Case to Case Basis
I		Condenser Fan Details		
	1	Make	-	Kirloskar Approved Vendor
	2	Fan Speed	RPM	910
	3	No. of Blades	Nos.	7
	4	Static Pressure	Pa	70
	5	Noise Level	dB	73
	6	Motor Details		
	i	Motor Type	-	3 Phase Induction Motor
	ii	Starter Type	-	DOL
	iii	Class of Insulation	-	Class F
	iv	Motor Rating	kW	1.4
	v	Motor Current	A	2.3
	vi	Supply Voltage	V	415
	vii	Phase	-	3
	viii	Frequency	Hz	50
	ix	Motor Protection Class	-	IP54
	7	Material of Construction		
	i	Fan Blades	-	Aluminum Alloy
	ii	Motor	-	Aluminum
	iii	Safety Guard	-	Steel Wire
J		Chiller Base Frame Details		
	1	Material	-	Contact To Engineering Team
	2	Method and details of construction OR Nature and joints used-folded/Welded /Bolted	-	Welded Bottom Frame and Remaining Components are Bolted with Bottom Frame
	3	Finish - Hot Dip Galvanised, Corrosion Resistant etc.	-	Spray Galvanising for MS Material NA for GI Material (Coat Base frame and Casing with a Corrosion-Resistant Coating Capable of withstanding a 1000 hour Salt-Spray Test According to ASTM B117)
K		Suction Line		
	1	Design Code	-	ASME B31.3
	2	Isolation Valve	-	Butterfly Valve
	3	Material of Construction	-	Carbon Steel
	4	Material Standard	-	Refer "MOC" Sheet
	5	Angle Valve	-	Provided on Suction Line For Service Purpose
L		Discharge Line		
	1	Design Code	-	ASME B31.3
	2	Isolation Valve	-	Shut-off Valve
	3	Material of Construction	-	Carbon Steel
	4	Material Standard	-	Refer "MOC" Sheet
	5	Skin Type Thermowell	-	Provided on Discharge Line For Discharge Temp. Sensor
M		Liquid Line		
	1	Design Code	-	ASME B31.3
	2	Expansion Valve		
	i	Type	-	Electronic Expansion Valve
	ii	Make	-	Kirloskar Approved Vendor
	iii	Quantity	Nos.	Contact To Engineering Team
	iv	Sight Glass	-	Inbuilt
	v	Moisture Indicator	-	NA
	3	Filter Drier	-	Provided
	4	Material of Construction	-	Copper
	5	Material Standard	-	Refer "MOC" Sheet
N		Desuperheater		
	1	Type	-	Plate Type
	2	Quantity	Nos.	One per Compressor
	3	Operating Conditions		
	i	Heat Duty	kW	Depends on Working Conditions
	ii	Hot Water Inlet Temperaure	°C	Depends on Site Conditions (Max. Possible - 55)
	iii	Hot Water Outlet Temperaure	°C	Max. Possible - 60

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		iv	Hot Water Flow Rate	L/s	Depends on Working Conditions
	4		Material of Construction	-	Brazzed PHE, Plate Material - SS
	5		Water Side End connection Details		
			Water Inlet Connection	NB	Consult with Engineering Department on Case to Case Basis
			Water Outlet Connection	NB	Consult with Engineering Department on Case to Case Basis
	6		Pressure Drop		
		i	Water Side	bar	less than 0.5
		ii	Refrigerant Side	bar	Proprietary Data
O			Economizer		
	1		Type	-	Plate Type
	2		Quantity	Nos.	One per Compressor

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	3	Heat Duty	kW	Proprietary Data
	4	Material of Construction	-	Brazed PHE, Plate Material - SS
P		Starter and Control Panel		
	1	Panel Enclosure	-	Starter and Control Panel Integrated in Single Fabricated Box
	2	Make	-	Kirloskar Approved Vendor
	3	Material of Enclosure	-	Fabricated Enclosure - GI
	4	Thickness of Enclosure	mm	Fabricated Enclosure Load Bearing Member - 2 mm Non-Load Bearing Member - 1.6 mm
	5	Ingress Protection (IP)	-	Consult with Engineering Department on Case to Case Basis
	6	Painting Specification		
	i	Paint Type	-	RAL 7035
	ii	Standard	-	Coating as per KCPL Standards
	7	Mounting Arrangement	-	Mounted on Chiller
	8	Type of Starter	-	Star-Delta Starter (Soft Starter / VFD - Optional)
	9	Type of Isolation	-	MCCB in case of Star-Delta Starter FSD in case of Soft Starter Consult with Engineering Department on Case to Case Basis in case of VFD Starter
	10	Type of Protection	-	MCCB in case of Star-Delta Starter FSD in case of Soft Starter Consult with Engineering Department on Case to Case Basis in case of VFD Starter
	11	Switchgear Make	-	Siemens
	12	Electrical and Control Cables	-	Power - PVC Insulated Single Core (Vtg. Grade 1.1 kV) Control- PVC Insulated Single Core, Multicore Cable (Vtg. Grade 1.1 kV) Signal- Shielded Cable
	13	Optional Features		
	i	Phase Indicating Lamps	-	Special-Optional
	ii	Hooter	-	Special-Optional
	iii	Energymeter	-	Special-Optional
	iv	Door Handle	-	Special-Optional
	v	LOTO Arrangement	-	Special-Optional
	vi	VFD for Condenser Fans	-	Special-Optional
Q		Controller		
	1	Make	-	Refer "Make List" Sheet
	2	Transmitters	-	NA
	3	Oil Level Switch	-	NA
	4	Oil Level Failure Trip	-	NA
	5	LP Switch and Gauge	-	No, Controller Program will Take Care of Low Pressure
	6	HP Switch and Gauge	-	No, Controller Program will Take Care of High Pressure
	7	Chilled Water Flow Failure	-	Yes
	8	Cooling Water Flow Failure	-	Yes
	9	Reverse Rotor Protection	-	No
	10	High/Low Voltage Trip	-	Yes
	11	Low Current Trip (Current Based-Analog)	-	Yes
	12	High Current Trip (Current Based-Analog)	-	Yes
	13	Phase Failure/Reverse Phasing Trip	-	Yes
	14	Earth Fault Trip	-	No
	15	Communication Through RS232/RS485	-	RS485
	16	Display of Microprocessor	-	Yes
	17	Type of Display	-	PGD0 Screen
	18	Remote Monitoring Facility	-	Yes
	19	Output to DCS	-	Applicable (Only if RS485 is Available)